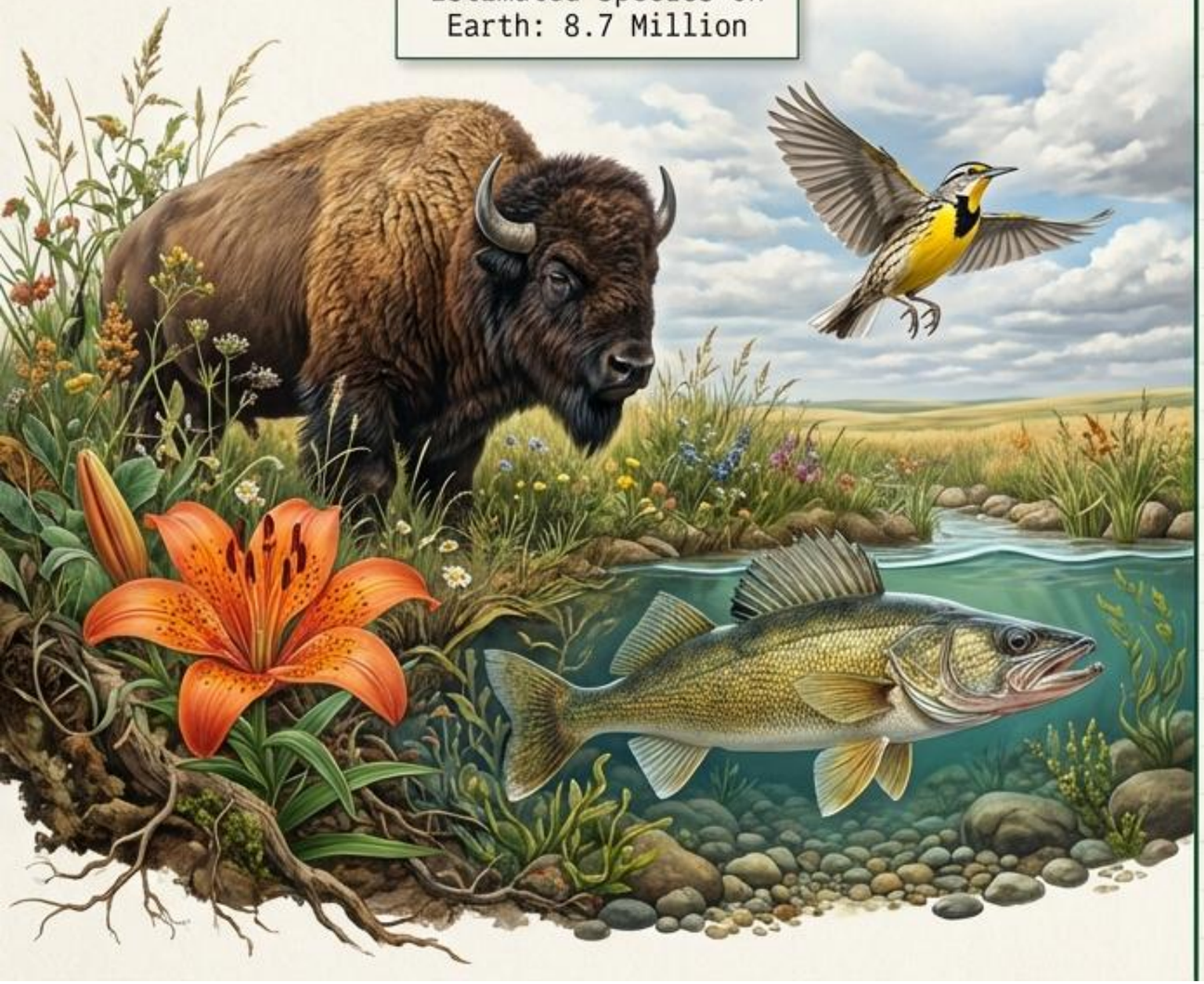
BIOLOGY 30: THE ORGANIZATION OF LIFE

Chapter 3 Review: Biological Classification

Estimated Species on Earth: 8.7 Million



Officially Named: 1.2 Million

SPECIMEN: Bison bison	
COMMON NAME: American Bison	
CONSERVATION STATUS: Near Threatened	
BARCODE:	DATE: OCT 26, 2024

SPECIMEN: Lilium philadelphicum	
COMMON NAME: Western Red Lily	
FAMILY: Liliaceae	FLOWERING PERIOD: June - July
BARCODE:	LOCATION: SK, Canada

SPECIMEN: Sander vitreus	
COMMON NAME: Walleye	
ORDER: Perciformes	LENGTH: 45-60 cm (avg)
BARCODE:	COLLECTION ID: 810-38-0001

SPECIMEN: Sturnella neglecta	
COMMON NAME: Western Meadowlark	
FAMILY: Icteridae	DIET: Insects, Seeds
BARCODE:	OBSERVER: A. Naturalist

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Nature doesn't come with labels.

Classification is not a universal truth. It is a human tool designed to make a staggering variety of life predictable and manageable.

Is this a weed or a crop?

Human Need: Survival/Economics

Does this have wings?

Human Need: Basic Observation

What is its DNA sequence?

Human Need: Evolutionary History

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Three ways to sort life

Functional Classification

Based on human utility (Artificial system).

Example: Grouping plants into flowers, vegetables, and weeds.

Morphological Classification

Based on physical appearance.

Example: Grouping bats and birds together because both possess wings (can be scientifically misleading).

Phylogenetic Classification

Based on evolutionary ancestry and DNA (Natural system).

Example: Grouping bats closer to mice than to birds based on genetic history.

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Case Study: The Weed Paradox

The Biological View (Natural System)

Genetically, Wild Oats are almost identical to the oats grown for food. They belong to the same biological family.

The Economic View (Artificial System)

Under the Saskatchewan Seeds Act, Wild Oats are classified as a Noxious Weed because they steal water from high-value crops.

Both classifications are correct. One serves **biology**; the other serves **agriculture** and **survival.**

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The Universal Language: Binomial Nomenclature

Invented by Swedish botanist Carl Linnaeus in the 1700s to solve regional confusion (e.g., a gopher in Saskatchewan is a ground squirrel, but in the US South, it's a tortoise).

Castor canadensis

Genus. Describes the group of closely related animals. Always Capitalized.

Specific Epithet. Identifies the exact species. Always lowercase.

Formatting Rules. Must be written in Latin (a stable dead language) and always italicized (or underlined if handwritten).

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Hierarchy in Action: *Vulpes vulpes*

Domain: Eukarya
(Cells have a nucleus)

Kingdom: Animalia
(Multicellular, eats food)

Phylum: Chordata
(Has a spinal cord)

Class: Mammalia
(Has fur, produces milk)

Order: Carnivora
(Sharp teeth for meat)

Family: Canidae
(Dog-like animals)

Genus: Vulpes
(True foxes with pointed ears)

Species: *Vulpes vulpes*
(The specific Red Fox)

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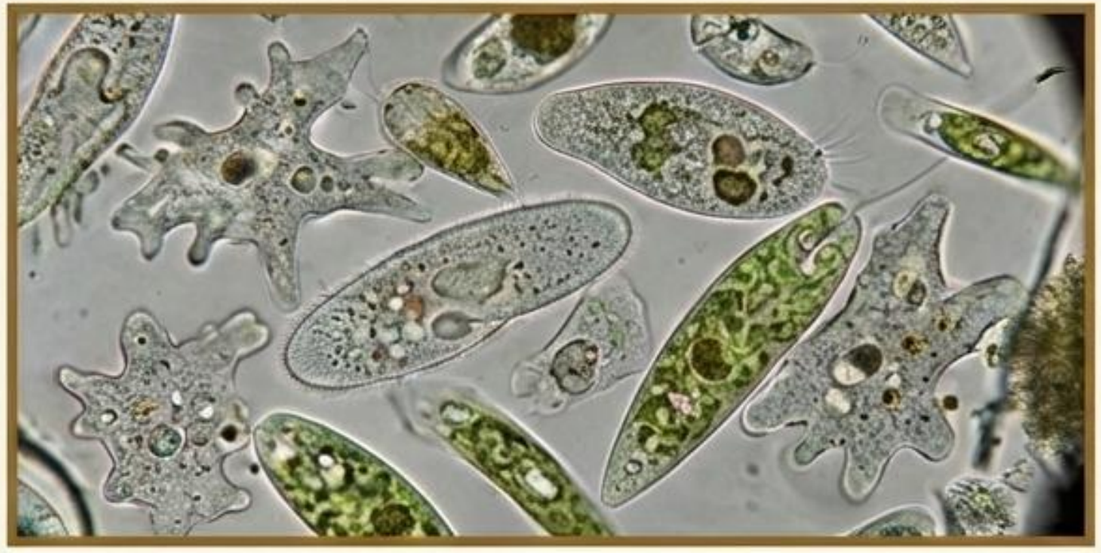
When Nature Resists the Boxes

The Junk Drawer



Subject: Kingdom Protista.

Issue: Single-celled organisms historically thrown here simply because they didn't fit perfectly as plants, animals, or fungi. DNA is now forcing us to reorganize them.

The Hybrid Breaker

Subject: The Pizzly Bear.

Issue: The Biological Species Concept states that separate species cannot interbreed to produce fertile offspring. Climate change is pushing Grizzlies and Polar bears together, creating fertile hybrids and breaking our rules.

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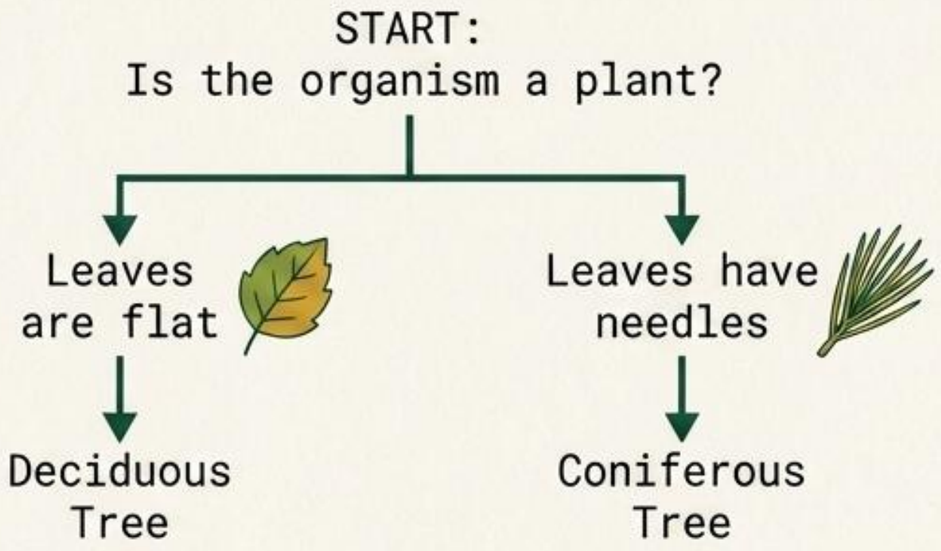
The Biologist's Toolkit

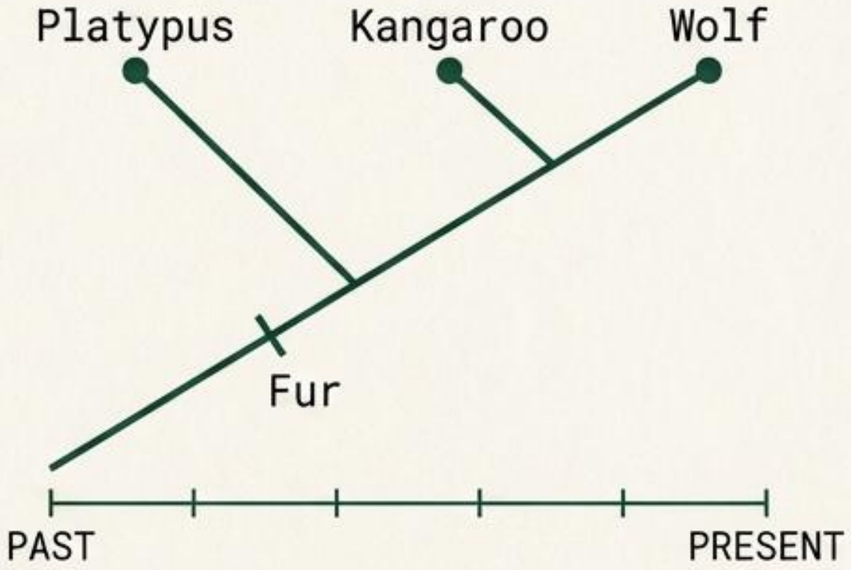
Dichotomous Key



Function: Tells you **WHAT** it is.

Mechanism: Uses **opposing physical traits** to identify unknown organisms.

Cladogram



Function: Tells you **WHERE** it came from.

Mechanism: Tracks **evolutionary history**. Each fork represents a **common ancestor**; markings along the branches show **derived characters** that evolved over time.

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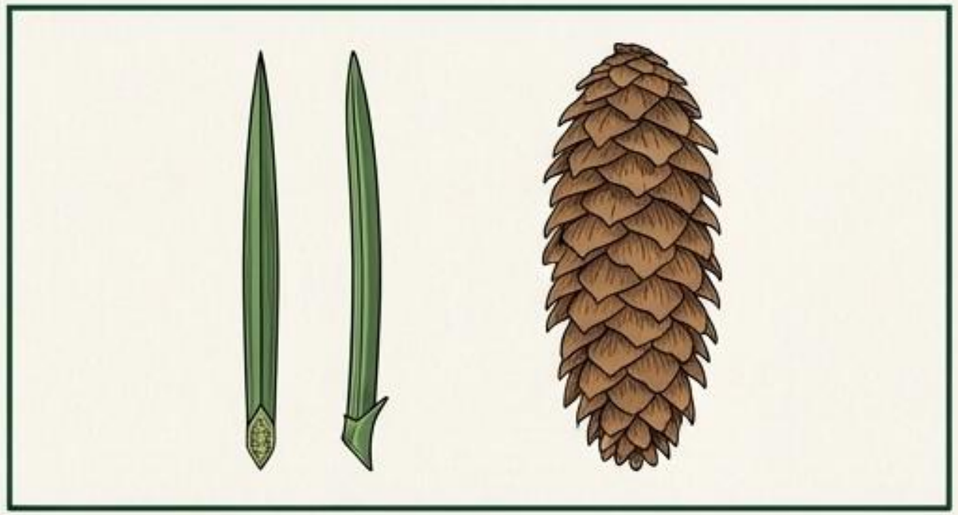
Prep: Conifer Identification Lab

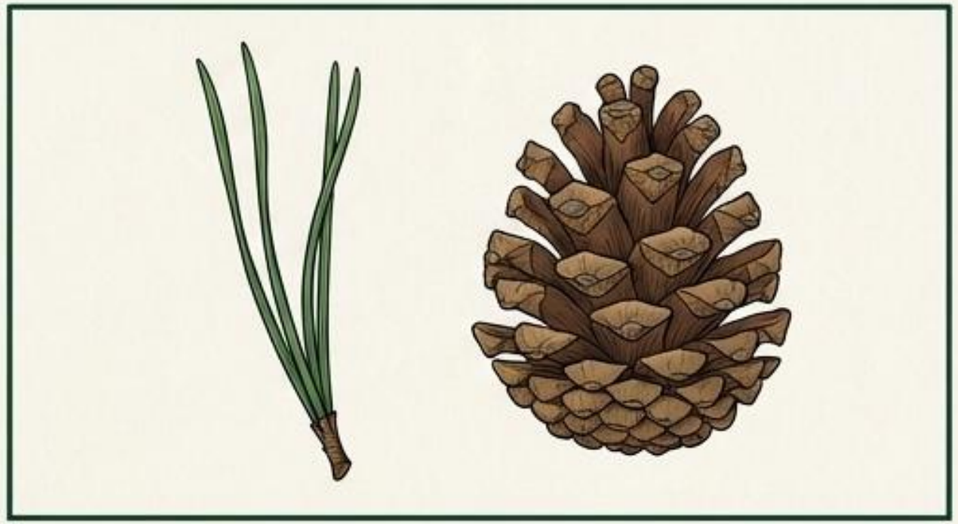
Mission:	Identify specific evergreen species in the Saskatchewan Boreal forest.
Method:	Utilize a dichotomous key to evaluate needle length, bundle structure, and cone shape.

Key Distinctions:

- **White Spruce:** Single, sharp needles.
- **Jack Pine:** Needles growing in bundles of two.



White Spruce (*Picea glauca*) - Needle & Cone



Jack Pine (*Pinus banksiana*) - Needle Bundle & Cone

Global Impact: Standardized naming means if you identify a Jack Pine here, a scientist in Sweden knows exactly what species you mean.

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Next Steps: Independent Study

Your 30-Minute Mission Parameters

- [ ] Review the structural differences between *Wâhkôhtowin* (circular kinship) and *Linnaean taxonomy* (hierarchical ranks).
- [ ] Practice correctly formatting *Binomial Nomenclature* (capitalization and italics) for local species.
- [ ] Complete the Interactive *Cladogram Lab* & the *Taxonomy Challenge Arcade Game* in your digital textbook.

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